

A COLLEAGUE KNOWLEDGE SHARE

Linear Models & Beyond

From regression lines to support vector machines — the core ideas behind regression, regularization, explainability, and classification.

Machine Learning Fundamentals • Regression & Classification

Two families of problems, one toolbox

01

Regression

Predicting a continuous number

- Linear models & linear regression
- Evaluation metrics (MAE, MSE, R^2)
- Ridge & Lasso regularization
- SHAP for explainability

02

Classification

Predicting a category

- Logistic regression & decision boundaries
- Support Vector Machines
- Kernels: linear, poly, RBF
- Evaluating classifiers

What are linear models?

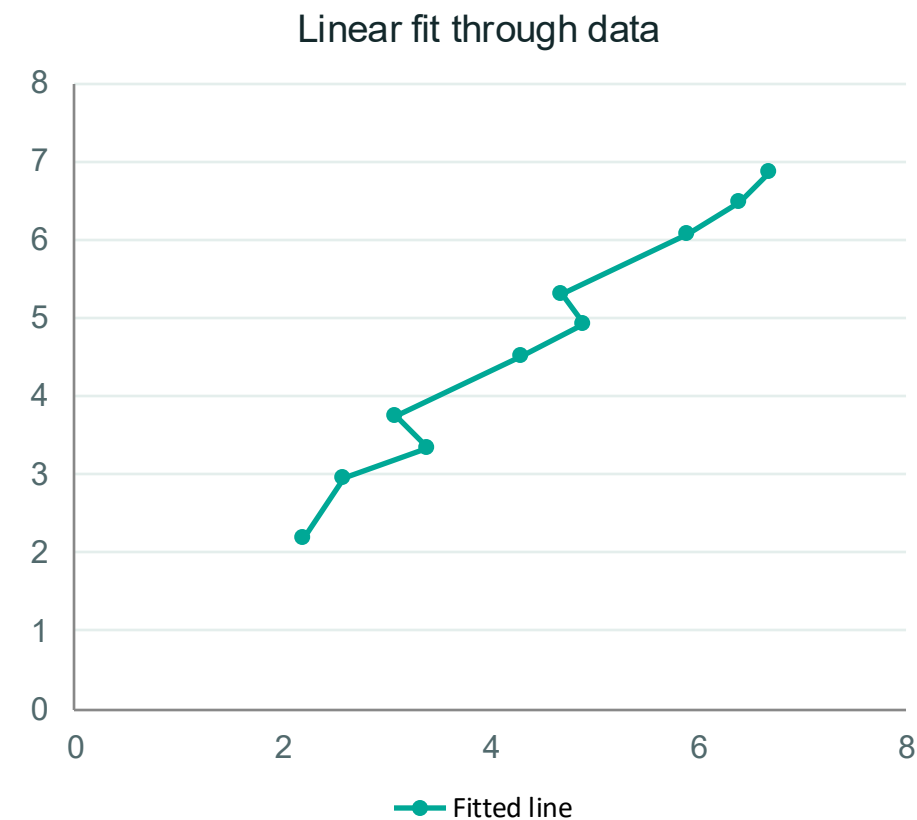


A model that assumes a straight-line relationship

Linear models predict an output as a weighted sum of input features, plus a bias term. Each feature gets a coefficient describing how much it moves the prediction.

$$y = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$$

weights (w) capture feature importance • bias (b) shifts the baseline



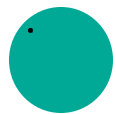
Linear regression: predicting a number

Linear regression finds the weights that minimize the gap between predicted and actual values — typically the sum of squared errors. It's the starting point for most regression work: fast, interpretable, and a strong baseline.



Forecasting

Sales, demand, revenue over time



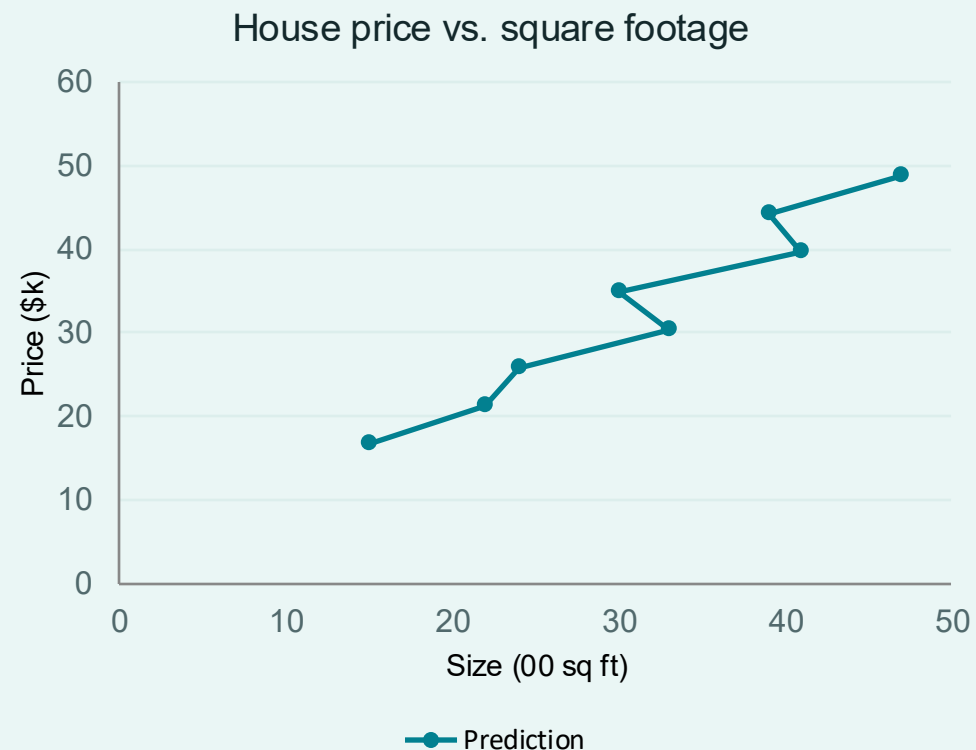
Pricing

Estimating house or product prices



Risk scoring

Continuous risk or cost estimates



How do we know a regression model is good?

MAE

Mean Absolute Error

Average size of the errors, in the original units. Easy to interpret, less sensitive to outliers.

MSE

Mean Squared Error

Squares each error before averaging — penalizes large mistakes much more heavily.

RMSE

Root Mean Squared Error

Square root of MSE — back in the original units, still outlier-sensitive.

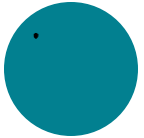
R²

Coefficient of Determination

$$R^2 = 1 - (\Sigma(y - \text{predicted } y)^2 / \Sigma(y - \text{mean } y)^2)$$

Share of the variance in the target explained by the model, from 0 to 1.

Ridge regression: taming overfitting

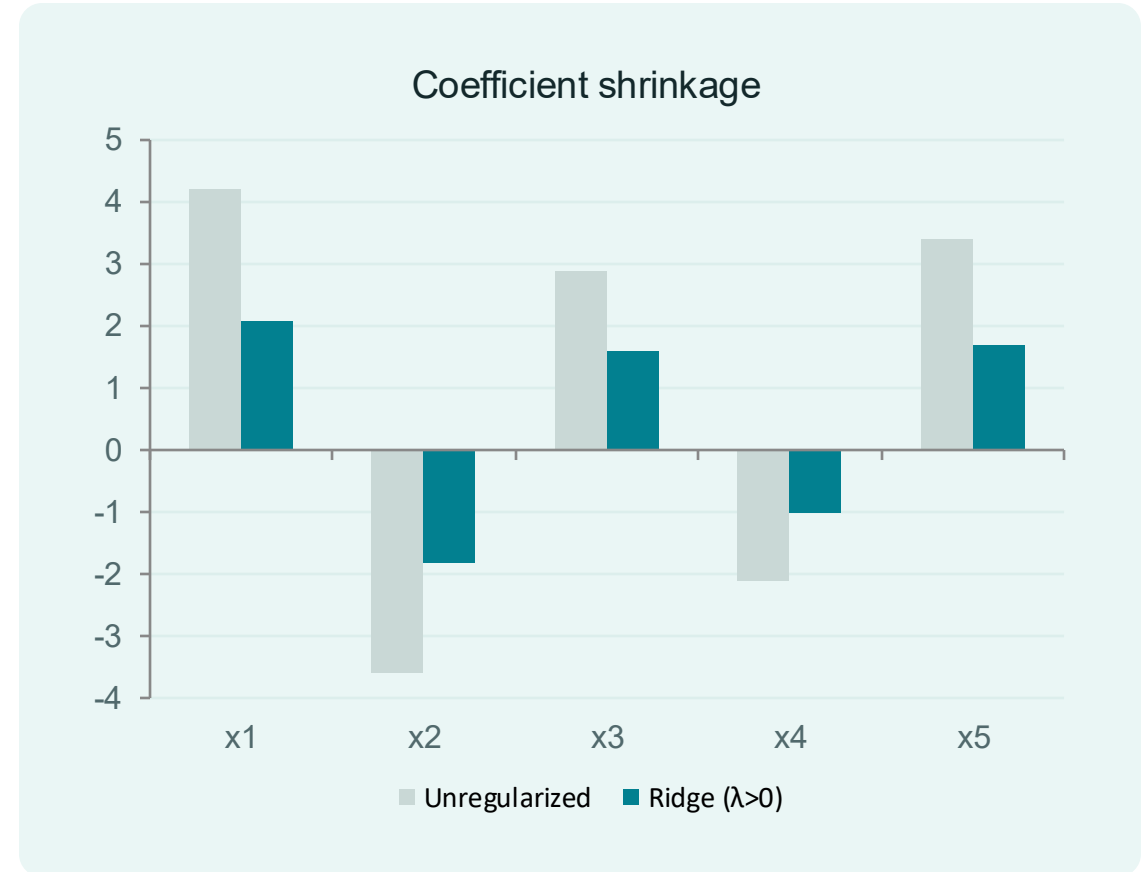


Ridge adds an L2 penalty — the sum of squared weights — to the loss function. Large coefficients get expensive, so the model shrinks them toward zero without ever eliminating them.

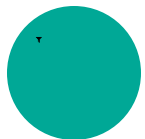
$$\text{Loss} = \Sigma(y - \hat{y})^2 + \lambda \cdot \Sigma w^2$$

λ controls the strength of the penalty

- Reduces variance from correlated / noisy features
- Keeps every feature in the model (weights shrink, never hit zero)
- Higher $\lambda \rightarrow$ simpler model, more bias, less overfitting



Lasso regression: built-in feature selection

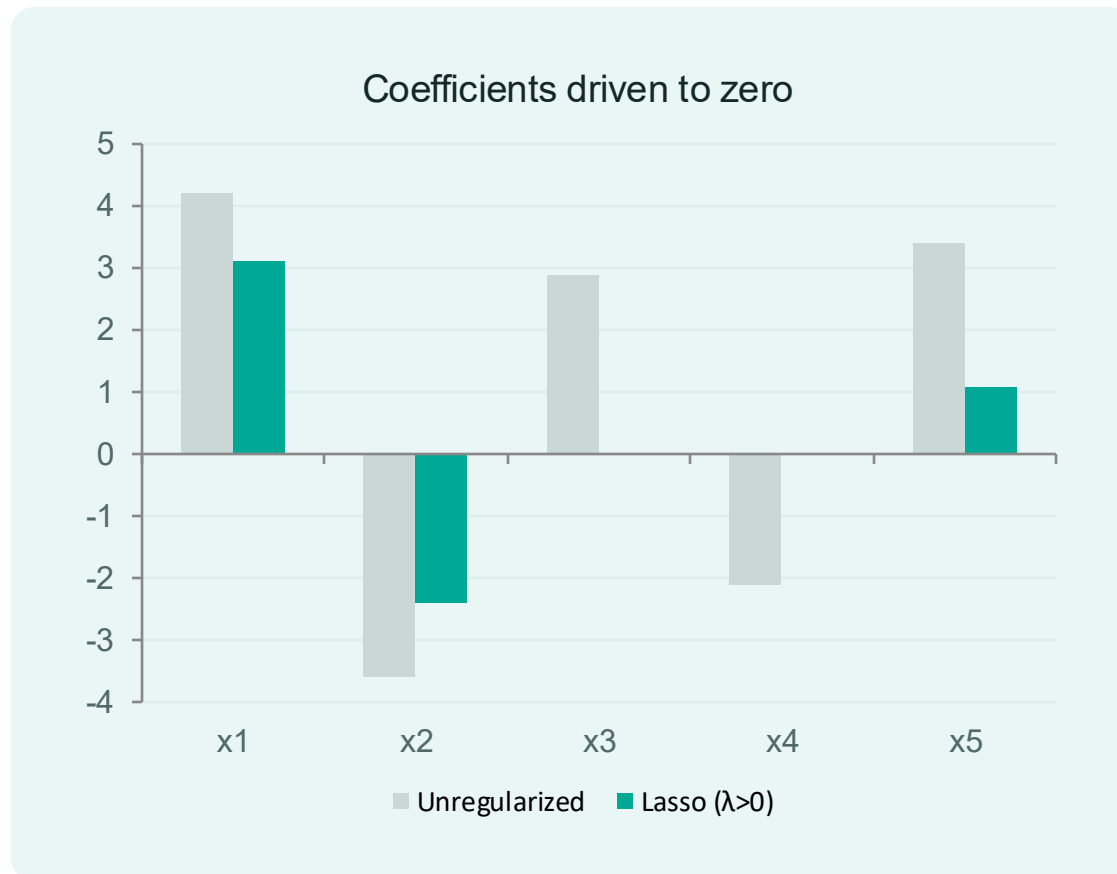


Lasso adds an L1 penalty — the sum of absolute weights. Unlike Ridge, this penalty can push weak coefficients all the way to exactly zero, effectively removing those features.

$$\text{Loss} = \Sigma(y - \hat{y})^2 + \lambda \cdot \Sigma|w|$$

The sharp corners of $|w|$ make zero an achievable minimum

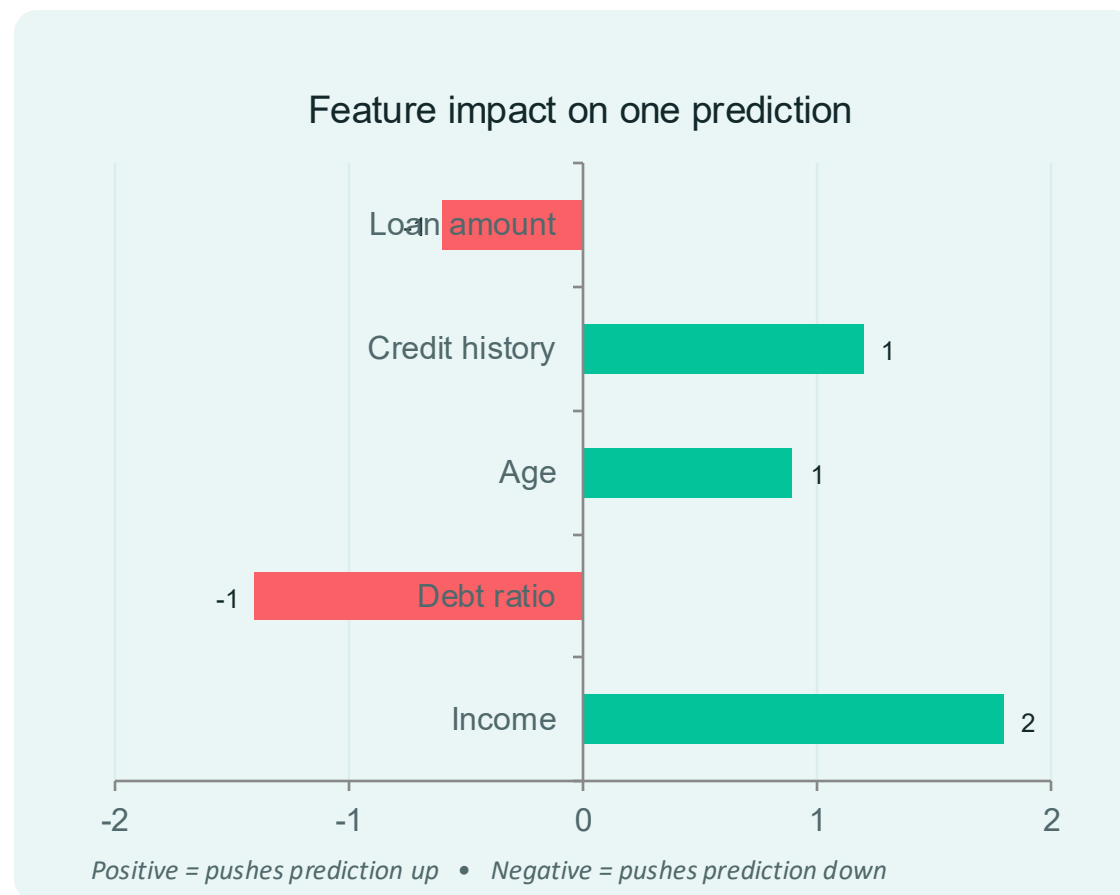
- Automatically drops uninformative features
- Produces sparse, more interpretable models
- Useful when you suspect only a few features matter



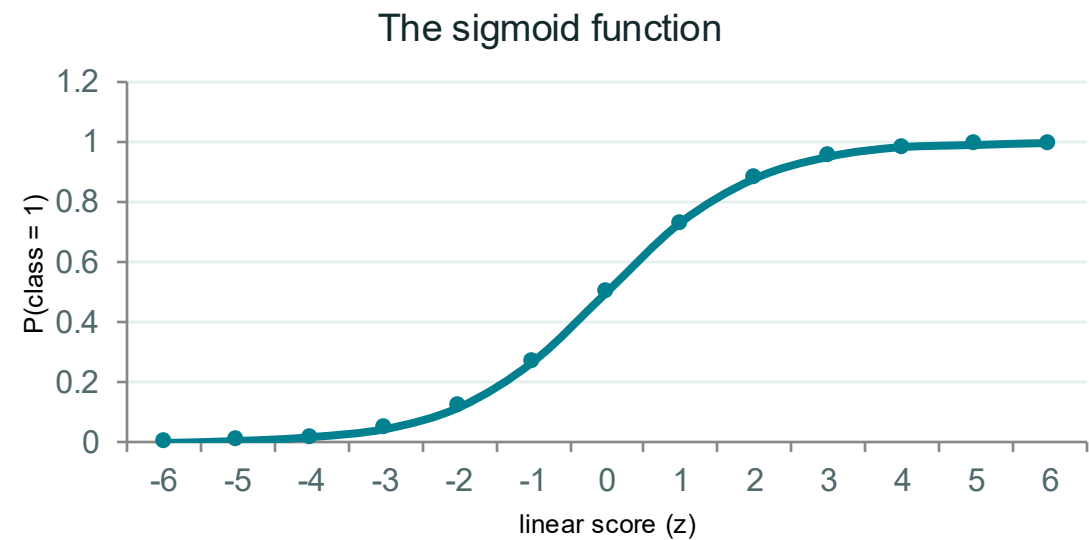
SHAP values: explaining a single prediction

SHAP (SHapley Additive exPlanations) borrows an idea from game theory: treat each feature as a “player” and fairly split credit for the prediction among them. Every feature gets a SHAP value — how much it pushed this one prediction above or below the average.

- Works on any model — linear, tree-based, or deep learning
- Local: explains one prediction, not just the model overall
- Values are additive: base value + SHAP values = prediction



Logistic regression: same idea, new job



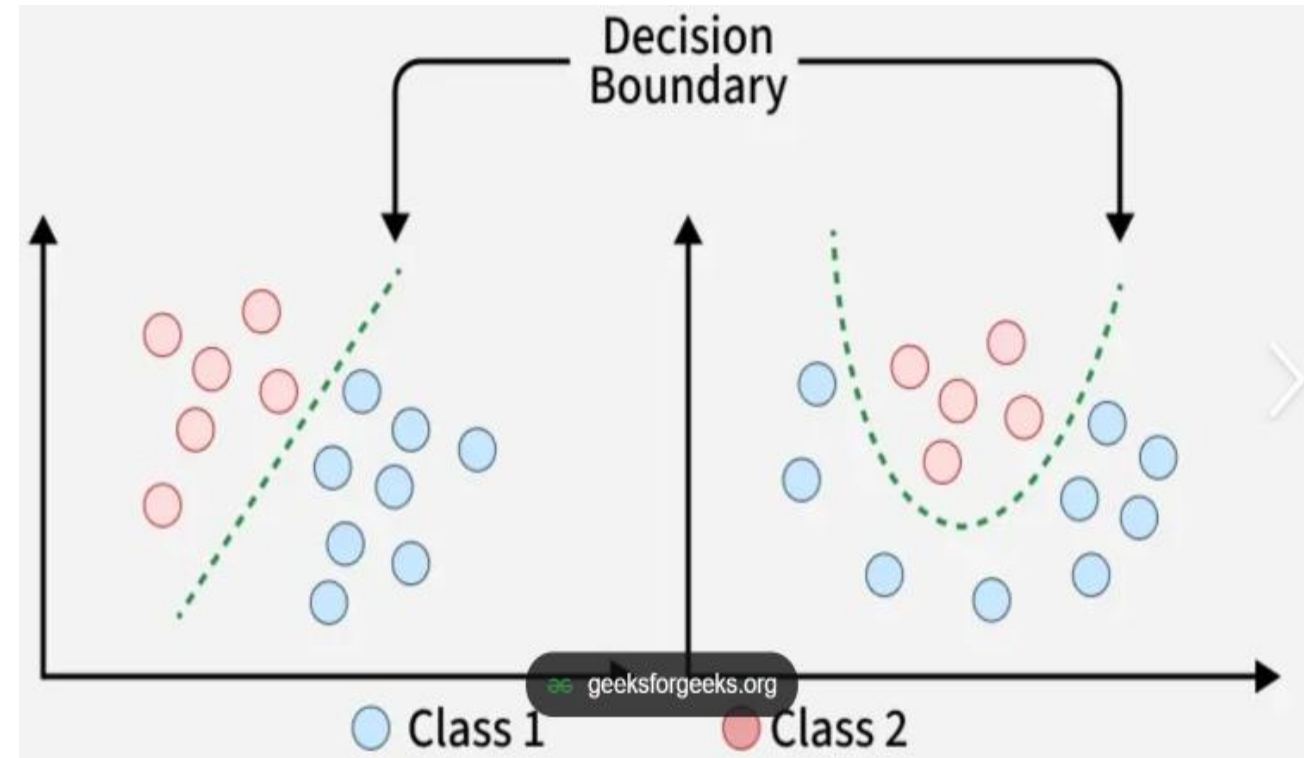
Logistic regression fits a linear score, then squashes it through the sigmoid into a probability between 0 and 1 — ready to threshold into a class.

	Linear regression	Logistic regression
Predicts	A continuous number	A class probability
Output range	$(-\infty, \infty)$	$(0, 1)$
Link function	None (identity)	Sigmoid / logit
Loss function	Squared error	Log loss (cross-entropy)
Typical use	Price, demand, scores	Spam? Churn? Default?

Decision boundaries: where the model draws the line

A decision boundary is the surface in feature space where the model switches its predicted class. Everything on one side is predicted class A; everything on the other is class B. For logistic regression the boundary is a straight line (or flat hyperplane); more flexible models can bend it.

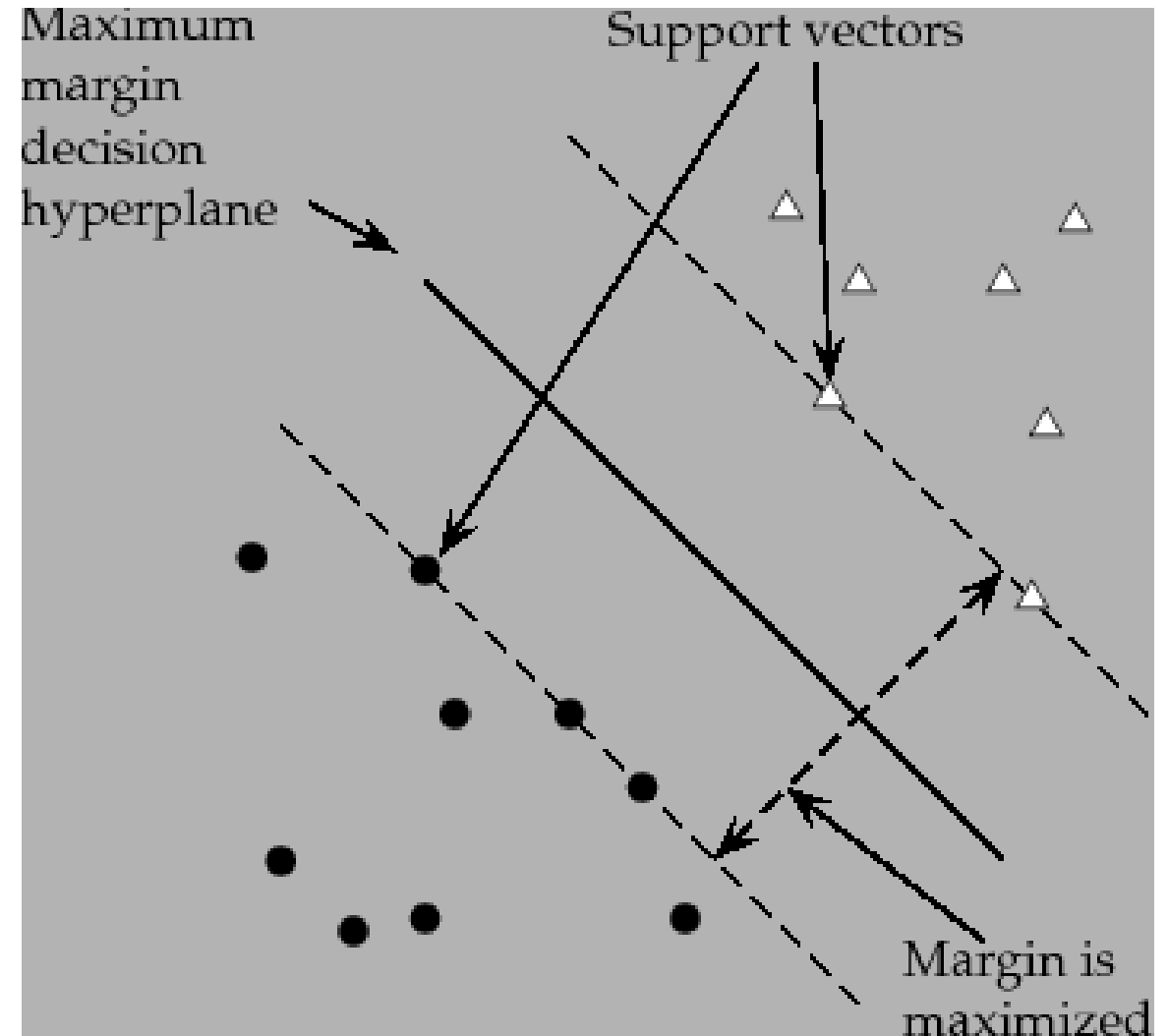
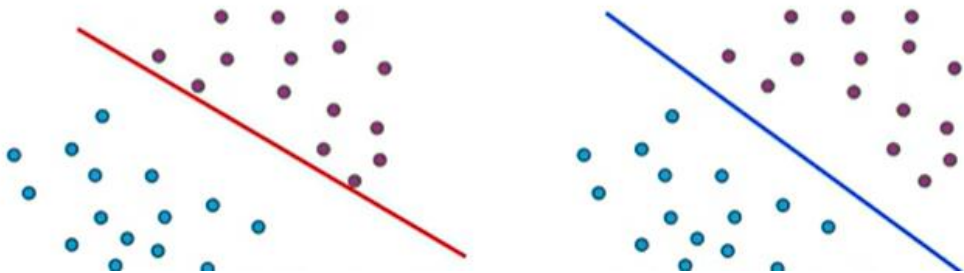
- Points are classified by which side of the boundary they fall on
- Boundary shape reflects model complexity — linear vs. curved
- Points near the boundary are the model's least confident cases



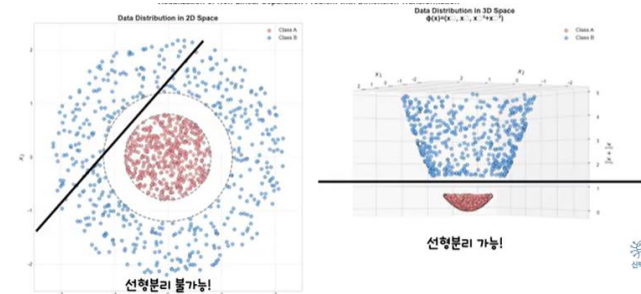
Support Vector Machines: maximizing the margin

An SVM doesn't just find a boundary — it finds the boundary with the widest possible margin to the nearest points of each class. Those closest points are the “support vectors”; they alone determine where the boundary sits.

- Maximizing the margin improves generalization to new data
- Only support vectors matter — other points can move freely
- A soft margin allows some misclassification for robustness



Kernels: letting SVMs bend the boundary



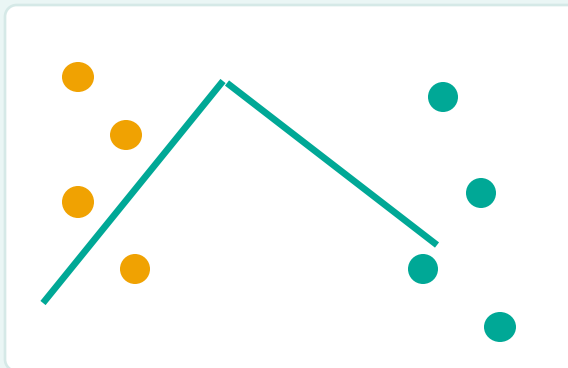
A kernel is a function that implicitly maps data into a higher-dimensional space, so a straight boundary there corresponds to a curved one back in the original space — without ever computing that mapping explicitly.

Linear



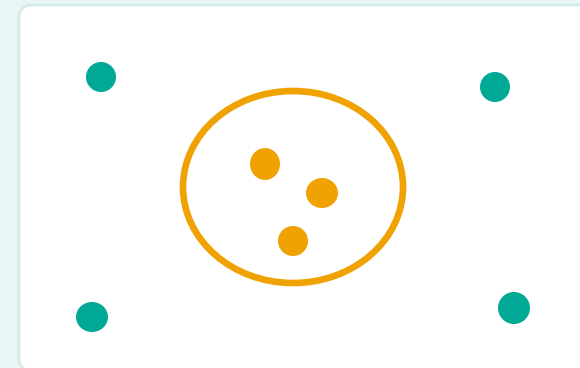
Straight boundary. Fast, interpretable, works well when classes are linearly separable.

Polynomial



Curved boundary of a chosen degree. Captures feature interactions, more flexible than linear.

RBF



Highly flexible, localized boundary. Good default for complex, non-linear patterns.

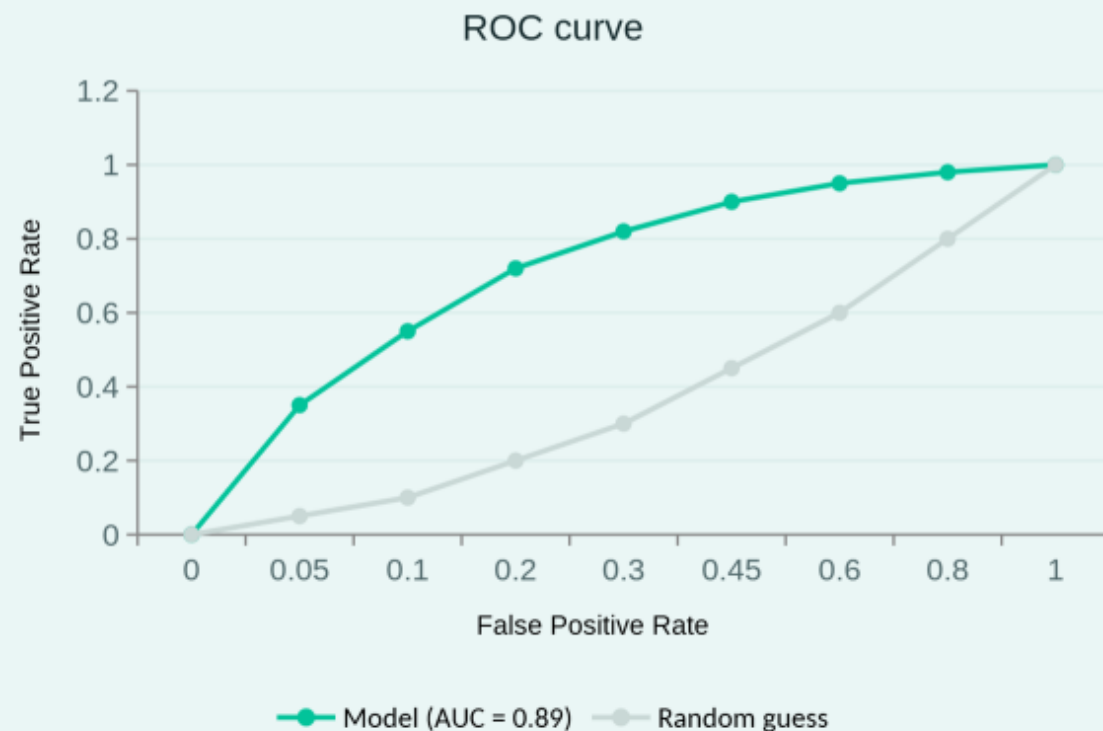
CLASSIFICATION

How do we know a classifier is good?

Confusion matrix

	Predicted +	Predicted -
Actual +	TP	FN
Actual -	FP	TN

Metric	Formula	Meaning
Accuracy	$(TP+TN) / \text{total}$	Overall correctness
Precision	$TP / (TP+FP)$	Of predicted positives, how many are right
Recall	$TP / (TP+FN)$	Of actual positives, how many were caught
F1-score	harmonic mean of P & R	Balance of precision and recall



KEY TAKEAWAYS

Bring it back to the team

1

Start simple

Linear and logistic regression are fast, interpretable baselines — reach for them first.

2

Regularize by default

Ridge and Lasso guard against overfitting; Lasso also does feature selection for free.

3

Explain predictions

SHAP values turn any model's output into a story stakeholders can trust.

4

Match the model to the boundary

Use SVM kernels when classes aren't linearly separable — RBF is a strong default.

5

Evaluate with the right metric

Precision, recall, and ROC-AUC matter more than accuracy alone when classes are imbalanced.